



## Exercise Sheet 7

### Exercise 1 — *Van der Waals interaction of graphene with Au and SiO<sub>2</sub>*

A graphene monolayer is located at a distance  $D$  above an atomically flat substrate. The effective van der Waals interaction energy (per unit area) between graphene and the substrate is described by

$$\frac{E(D)}{A} = \frac{K}{D^9} - \frac{C_3}{D^3},$$

with  $K > 0$  (repulsive term) and  $C_3 > 0$  (attractive vdW term). The exponent  $D^{-3}$  reflects the long-range van der Waals attraction between a 2D sheet and a bulk substrate. The much steeper  $D^{-9}$  repulsive term models short-range Pauli repulsion arising from overlapping electronic wavefunctions.

Two substrates are to be compared:

- Substrate 1: SiO<sub>2</sub> with van der Waals constant  $C_3^{(\text{SiO}_2)}$ ,
- Substrate 2: Au with a stronger interaction

$$C_3^{(\text{Au})} = 3 C_3^{(\text{SiO}_2)}.$$

The repulsive constant  $K$  is identical for both substrates.

- Determine the equilibrium distance  $D_0$ , at which  $\frac{E(D)}{A}$  is minimal, in terms of  $K$  and  $C_3$ .
- Determine the binding energy per area at the equilibrium distance,

$$\left. \frac{E(D)}{A} \right|_{D=D_0},$$

again expressed in terms of  $K$  and  $C_3$ .

- Using now  $C_3^{(\text{Au})} = 3 C_3^{(\text{SiO}_2)}$ , compute:

- The ratio of the equilibrium distances  $D_0^{(\text{Au})}/D_0^{(\text{SiO}_2)}$ .
- The ratio of the magnitudes of the binding energies  $\left| E_{\min}^{(\text{Au})}/A \right| / \left| E_{\min}^{(\text{SiO}_2)}/A \right|$ .

Comment qualitatively on what these results imply about the relative binding strength of graphene on Au compared to graphene on SiO<sub>2</sub> (e.g., in terms of adhesion and exfoliation).



**Exercise 2** — *One-dimensional ionic crystal (alternating  $\pm q$ )* [moderate]

Consider a chain of  $N$  ions (i.e.  $N/2$  pairs) at spacing  $R$  with alternating charges  $\pm q$ ; nearest-neighbour repulsion is  $A/R^n$  ( $n > 1$ ).

- (a) Compute the Madelung constant  $\alpha$  of the infinite chain.
- (b) Find the equilibrium spacing  $R_0$ .
- (c) Show that  $U(R_0) = -\ln 2 \frac{Nq^2}{4\pi\epsilon_0 R_0} \left(1 - \frac{1}{n}\right)$ .

**Exercise 3** — *Ionic crystal made of identical atoms* [moderate]

Imagine a crystal whose binding arises from the Coulomb attraction between the positive and negative ion of the *same* atom or molecule  $R$ . Such charge-transfer ion pairs occur in some organic molecules, but ionic crystals of the form  $R^+R^-$  are never observed when  $R$  is a single atom.

- (a) Compute the binding energy of a hypothetical  $\text{Na}^+\text{Na}^-$  ionic crystal, assuming that the atomic separation in this crystal is the same as in metallic sodium,  $R = 3.72 \text{ \AA}$ . Compare the resulting binding energy with the cohesive energy of metallic sodium  $E_B = -1.13 \text{ eV}$  per atom.
- (b) Compare the stability of the  $\text{Na}^+\text{Na}^-$  ionic crystal with that of a  $\text{Na}^+\text{Cl}^-$  ionic crystal, assuming that both crystals have the fcc structure.

Given data:

Ionization energy of Na:  $I = +5.14 \text{ eV}$ ,

electron affinities:  $A_{\text{Na}} = -0.78 \text{ eV}$ ,  $A_{\text{Cl}} = -3.61 \text{ eV}$ ,

Madelung constant of the fcc structure:  $\alpha = 1.747$ ,

lattice constant of NaCl:  $a = 5.6402 \text{ \AA}$ .